1. A firecracker explodes in reference frame S at t = 1.0 s. A second firecracker explodes at the same position at t = 3.0 s. In reference frame $S^{'}$, which moves in the x-direction at speed v, the first explosion is detected at $x^{'} = 4.0$ m and the second at $x^{'} = -4.0$.

A. What is the speed of frame $S^{'}$ relative to S?

B. What is the position of the two explosions in frame S?

2. At t= 1.0 s, a firecracker explodes at x = 10 m in reference frame S. Four seconds later, a second firecraker explodes at x = 20 m. Reference frame $S^{'}$ moves in the x-direction at a speed of 5.0 m/s. What are the position and times of these two events in frame $S^{'}$?

3. A newspaper delivery boy is riding his bicycle down the street at 5.0 m/s. He can throw a paper at a speed of 8.0 m/s. What is the paper's speed relative to the ground if he throws the paper (a) forward, (b) backwards, (c) to the side?

4. A baseball pitcher can throw a ball with a speed of 40 m/s. He is in the back of a pickup truck that is driving away from you. He throws the ball in your direction, and it floats toward you at a lazy 10 m/s. What is the velocity of the truck?

5. An out-of-control alien spacecraft is diving into a star at a speed of $\frac{1}{3}$ c. What speed relative to the spacecraft is the starlight approaching?

6. A starship blasts past the earth at $\frac{2}{3}$ c. Just after passing the earth, it fires a laser beam out the back of the starship. With what speed does the laser beam approach the earth?